Biomolecules as Thermodynamic Computers, Subtractive Ontology and the Informational Collapse: The case of the Bacterial Flagellar Motor

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Abstract

Traditional models of bioenergetics treat biological life as an additive process: organisms accumulate structural complexity and consume free energy to maintain states far from thermodynamic equilibrium. Zurvatronics is proposed as an inhibitory framework in which agency emerges not from added complexity, but from the systematic subtraction of informational entropy the topological "pruning" of non-functional stochastic trajectories from a system's conformational phase space. This paradigm shifts the fundamental teleology of molecular machines: they do not compute in order to move; they compute to persist. The macroscopic mechanical work observed the directed rotation of the flagellum is considered the byproduct of this continuous informational inhibition. Drawing on Shannon information theory and Richard Swineshead's medieval intensity calculus (*latitudo formarum*, ca. 1350), the MotAB flagellar stator of *Pseudomonas putida* is modeled as an informational anchor. The Zurvān Coefficient (Z) quantifies cumulative filtering energy per proton transit, and the Swineshead Latitudo (L) measures the stator's instantaneous filtering capacity. While *P. putida* is ecologically classified as psychrotolerant maintaining chemotactic motility across 277–303 K its single-motor torque-speed relationship remains uncharacterized at the biophysical level. Bead-assay single-motor measurements are currently underway, with preliminary observations consistent with Zurvatronic predictions. The framework predicts that cold environments optimize informational pruning a "Zurvān Gap" rather than impeding motor function, and that the near-zero-load controversy in flagellar motor biophysics represents a predictable zone of informational instability rather than experimental error. This subtractive ontology offers design principles for synthetic molecular motors under thermodynamic constraint, with implications for living and lifelike complex adaptive systems.

Submission type: **Full Paper**, Data/Code available at:

https://github.com/Glyxon/Zurvatronics

Introduction

Contemporary bioenergetics models life as an additive process: molecular machines such as the bacterial flagellar motor (BFM) are understood as biological heat engines that convert proton motive force (PMF) into mechanical work against viscous drag (Purcell, 1977; Berry & Berg, 1997). This framework has been extraordinarily productive but encounters a persistent explanatory gap: It cannot fully account for the precision and persistence of molecular function in thermally fluctuating, low-temperature environments where available free energy approaches classical Arrhenius activation limits.

A parallel thread in biophysics suggests that biological systems actively exploit quantum coherence and decoherence as computational resources. Lambert *et al* (2013) formalized the concept of "quantumness" as a biological resource harnessed at the boundary between quantum coherence and classical decoherence not by isolating systems from their thermal bath, but by coupling dissipation to precisely engineered molecular geometries. Evidence for non-trivial quantum effects in photosynthetic energy transfer (Engel *et al*.,2007) and avian magnetoreception (Ritz et al., 2000) has established quantum biology as a legitimate and experimentally verified discipline.

Zurvatronics is proposed as an inhibitory theoretical framework in which biological agency arises from the systematic subtraction of informational entropy the active "pruning" of non-functional stochastic trajectories from the system's phase space. Under this ontology, biomolecules are not passive catalysts but thermodynamic computers whose primary computational act is the selective suppression of decoherent states. This reframing aligns with noise-filtering and kinetic proofreading literature (Hopfield, 1974; Eldar & Elowitz, 2010) while extending these concepts into molecular topology as an informational operator. The philosophical substrate of the framework draws from Zurvān the Zoroastrian principle of infinite time and primordial potentiality. There is a critical distinction between Zurvān Akarana (Infinite Time: the unstructured reservoir of all possible states, analogous to maximum entropy) and Zurvān Dareghō-chvadhāta (Time of Long Dominion: the stabilized, directed temporality characteristic of living systems). Life, under this reading, does not merely exist within time — it produces directed time by collapsing infinite stochastic potentiality into discrete, functional classical events.

The Zurvān Coefficient quantifies the efficiency of this collapse. The historical connection to Swineshead's latitudo formarum is precise rather than decorative: Swineshead's intensity calculus measures the actualization of a quality relative to its maximum potential, exactly what the Latitudo does for conformational precision.

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This work relies on the deployment of 14th-century logic as a thermodynamic measure grounded in Shannon information theory. As a paradigm case, the MotAB flagellar stator of *Pseudomonas putida* the psychrotolerant soil bacterium that maintains chemotactic motility across 277–303 K is analyzed. Its single-motor torque-speed behavior across this thermal range has not previously been characterized at the biophysical level. Single-motor bead-assay measurements are currently underway, with the Zurvatronic framework providing the predictive scaffold against which these experiments will be interpreted.

II-MATHEMATICAL FORMALISM

II.1 The Swineshead Latitudo *L(T)*

Instantaneous Filtering Capacity. The Latitudo is defined as the instantaneous thermodynamic energy invested by the stator's physical topology to maintain its conformational restriction:

$$L(T) = k_B T \ \ln 2 \cdot [H_{max} - H(S)] \quad \text{(Joules)} \qquad (1)$$

Where k_B is Boltzmann's constant, T is temperature, $H_{max} = log2(n)$ is the maximum Shannon entropy of n equally probable conformational states (bits), and $H(S,T)$ is the actual Shannon entropy of the stator's conformational distribution (bits). The factor $kBT{\cdot}ln2$ is Landauer's limit, the minimum thermodynamic energy to erase one bit of information (Bérut et al., 2012). Multiplying by the entropy reduction $[Hmax - H(S,T)]$ yields Joules.

Physical interpretation: $L=0$ when the stator is completely disordered (no filtering capacity); L is maximized when the stator occupies a single conformational state (perfect filter). The temperature dependence is non- trivial: as T decreases, the $k_B T$ prefactor decreases linearly, but $H(S,T)$ decreases as fewer conformational states become thermally accessible, so $[Hmax - H(S,T)]$ rises steeply. It is modeled as:

$$H(S,T) = H_{\max}\left(1 - e^{-\left(\frac{T}{T_0}\right)^2}\right) \qquad (2)$$

Where T0 is a characteristic conformational temperature scale. The net result is that $L(T)$ peaks in the psychrotrophic range — cold environments provide higher filtering precision per unit thermal energy. This is the mathematical origin of the Zurvān Gap.

Connection to Swineshead: $[H_{max} - H(S,T)]$ is the direct analog of Swineshead's *intensio* the degree to which a quality (here, conformational precision) is actualized relative to its maximum potential. The *Latitudo* translates the medieval intensity calculus into a thermodynamically grounded energy metric.

II.2 The Zurvān Coefficient Z

The nomenclature of the Zurvān Coefficient (Z) is drawn from classical Zoroastrian philosophy (described above), specifically the concept of *Zurvān* as the primordial, unformed substrate of infinite time and space from which directed agency ultimately emerges (Zaehner, 1955). In this biophysical framework, Z represents the temporal accumulation of informational action required for the stator to manifest functional identity out of the stochastic void.

Cumulative Filtering Energy Z is the time-integrated filtering capacity of the stator over the proton's residence time τ_{res} within the MotAB channel:

$$Z = \int_{\theta}^{\tau_{\text{res}}} L(T,t)\, dt \qquad \text{(Joules/seconds)} \qquad (3)$$

Physical interpretation: Z represents the total informational work performed by the stator during a single proton transit event. Its units are Joules/seconds, the dimensions of action, identical to Planck's constant $\hbar$. This is not coincidental: it creates a non-trivial bridge to the quantum biological foundations of the framework, as the stator accumulates an "action-like" mass of filtering energy during each discrete proton transit.

Critical consequence:

As rotational speed increases, $\tau_{res} \to 0$, so $Z \to 0$

regardless of how large $L(T)$ is. The stator does not fail mechanically at high speed it runs out of time to perform sufficient informational work.

II.3 The Critical Threshold *Zc*

$$Z_c = k_{\text{B}} T_{\text{ref}} \ln(2)\left[H_{\max} - H(S, T_{\text{ref}})\right] \tau_{\min} \qquad \text{(Joules/seconds).} \qquad (4)$$

Where τ_{min} is the minimum temporal window required to process sufficient informational entropy to activate the stator's catch-bond (anchoring to the peptidoglycan), and T_{ref} is a reference temperature.

Within this subtractive framework, Z_c functions as an informational activation barrier that governs both single-cycle kinematics and population-level stability. During a standard proton transit, if the accumulated action $Z(t)$ reaches Z_c, the intrinsically disordered region (IDR) undergoes a functional phase transition (*stick-slip* collapse). This transition releases the restrictive coupling friction, generating a power stroke while securely maintaining the stator's anchored state.

However, if the rotational speed becomes faster (e.g., at near-zero load), the proton residence time τ_{res} drops below τ_{min}. Consequently, the accumulated action fails to reach the critical threshold ($Z<Z_c$) before the transit concludes.

Without processing this minimum informational requirement, the catch-bond mechanism fails to trigger, leading to stochastic disassembly of the stator unit from the motor complex (rapid fluctuations).

II.4 The Immanence Coefficient ζ —Population-Level Anchoring

$$\zeta = \frac{1}{N}\sum_{j=1}^{N}\frac{Z_j}{Z_c} \text{ (dimensionless)} \quad (5)$$

Where N is the number of actively engaged stator units. $\zeta > 1$ corresponds to a fully anchored, agency-sustaining motor; $\zeta < 1$ corresponds to the stochastic regime where informational pruning is insufficient and disassembly becomes probable. This sigmoidal transition around $\zeta = 1$ is the physical realization of the Adjacent Possible threshold (Kauffman, 2002).

II.5 The Inhibitory Operator Ψ_Z, Modified Langevin Dynamics

To interface the information-theoretic model with classical biophysics, Ψ_Z is introduced as an additive counter-force in the Langevin equation. However, to capture the threshold-dependent *stick-slip* nature of the motor's power stroke, this restoring force must be gated by the structural integrity of its intrinsically disordered region (IDR):

$$m\frac{dv}{dt} = F_{\text{PMF}} - \eta v - \left[P_{\text{assembly}}(\tau, T)\cdot \Psi Z(t)\right] + \xi(t) \quad (6)$$

Where the information penalty $\Psi Z(t)$ is defined as:

$$\Psi Z(t) = \kappa \cdot Z(t)\cdot \tau_{\text{res}} \text{ (Newtons)} \quad (7)$$

Here, κ is a phenomenological elastic coupling constant ($N{\cdot}J{-}1{\cdot}s{-}1$) that maps the accumulated informational action $Z(t)$ into a mechanical restoring force. Crucially, the force is directly proportional to the residence time τ_{res}, ensuring that rigid, slow-relaxing conformations impose greater informational braking on the rotor than rapid, fluid conformations. $P_{assembly}$ is the probability of the stator maintaining its active conformation, modeled as a sigmoidal function centered on the critical threshold Z_c . By gating the informational penalty through this probability, the Generalized Langevin Equation captures emergent agency: When the accumulated informational constraint Z exceeds Z_c, the conformation collapses ($P_{assembly} \rightarrow 0$), the restoring force vanishes, and the system yields the discrete acceleration burst characteristic of the motor's power stroke. The stability of a Zurvatronic agent is thus earned continuously through informational work and reset through thermal collapse, not guaranteed by static structural rigidity.

III. EMPIRICAL CASE STUDY: THE MotAB STATOR AS A STOCHASTIC WEDGE

III.1 The MotAB Stator as Stochastic Wedge

Within the Zurvatronic framework, the MotAB stator functions as a Stochastic Wedge: a molecular structure whose transmembrane topology actively narrows the accessible conformational phase space of the proton translocation pathway, suppressing non-functional trajectories and allowing only those consistent with productive rotor displacement to persist. The wedge metaphor is precise: like a physical wedge, the stator does not add energy to the system but constrains geometry, redirecting existing forces into productive directions.

Proton transport occurs via the Grotthuss mechanism rapid, non-diffusive hopping across a hydrogen-bond network within the stator channel (Agmon, 1995; Wraight, 2006). In the Zurvatronic framework, this high-speed relay functions as the informational sampling clock of the Stochastic Wedge. It provides the temporal resolution necessary for the stator to continuously update its conformational filtering. When the Grotthuss chain is intact, τ_{res} is sufficient for Z to accumulate beyond Zc, sustaining stator anchoring. When disrupted by mutation, thermal denaturation, or kinematic overload Z collapses and disassembly follows.

III.2 Empirical Support: The MotB 72-100 IDR

Critical supporting evidence comes from the intrinsically disordered region (IDR) of MotB spanning residues 72–100. Castillo *et al.,* (2013) demonstrated that in-frame deletion of this 28-amino acid segment produces a specific phenotype: wild-type torque at high load (low speed), but premature torque drop-off as load decreases and rotational speed increases. The motor is not paralyzed it is selectively impaired in the high-speed regime. The Zurvatronic interpretation: the MotB 72-100 IDR functions as a Functional Stochastic Manifold whose conformational flexibility extends the effective τ_{res} at high rotational speeds, preventing premature Z collapse. Its disorder is not a structural deficiency it is a computational requirement. In its absence, the motor enters informational blackout at high speeds: the proton transits the channel before sufficient filtering energy accumulates, Z drops below Z_c, and stator units disassemble. The torque-speed drop-off is not mechanical failure. It is the signature of a computation that has run out of time.

III.3 Elucidating the Zero-Load Controversy

A primary test of any novel framework is its capacity to resolve standing paradoxes. Nirody et al. (2017) demonstrated that motor speed near zero torque increases linearly with stator number—implying a low duty ratio and frequent stator disengagement at zero load. Concurrently,

Wang et al. (2017) used optical trapping to demonstrate that zero-load speed is entirely independent of stator number—implying a high duty ratio close to 1. Nakamura et al. (2020), probing near-zero load with 60 nm gold nanoparticles, observed massive discrete speed fluctuations and prolonged pausing events.

The Zurvatronic resolution is that this contradiction is not an artifact of experimental error, but the signature of a complex adaptive system operating precisely at the boundary of informational collapse ($\zeta \approx 1$).

As external load approaches zero, rotational speed reaches its maximum, pushing the proton residence time τ_{res} toward τ_{min} . Consequently, the accumulated action Z drops precipitously toward the critical activation barrier Z_c . At this edge of chaos, the stator's IDR is computing at its absolute thermodynamic limit.

Because the system is hovering on the sigmoidal boundary of $P_{assembly}$, minute differences in experimental conditions dictate categorically different macrodynamics. The restoring force of Wang's optical trap artificially injects a marginal increase in drag, extending τ_{res} just enough to keep $Z \geq Z_c$, resulting in a fully assembled motor (high duty ratio). Nirody's rotational shear assay lacks this drag, pushing τ_{res} slightly lower, forcing $Z < Z_c$. Here, stators fail to trigger their catch-bonds and continuously fall off, creating the linear dependence on stator number. Finally, the extreme fluctuations observed by Nakamura *et al*., are the direct, unmasked macroscopic signatures of individual stator units stochastically crossing the Z_c threshold in real-time.

The near-zero load regime is therefore not a zone of mechanical wear; it is a zone of fundamental informational instability. The motor does not break; it runs out of time to compute its own existence.

III.4 Experimental Status and Dual-Stator Complexity

Single-motor bead-assay measurements of *P. putida* KT2440 are currently underway using tethered-cell and flagellar stub configurations across the 277–303 K thermal range. Preliminary observations are consistent with Zurvatronic predictions. However, it is explicitly acknowledged that a critical complexity exists. Unlike *E. coli* or *Salmonella*, *P. putida* possesses two genetically independent stator systems—MotAB and MotCD—with functionally divergent roles (Pfeifer et al., 2022).

Within the Zurvatronic framework, these two stator systems are not merely redundant mechanical engines; they are thermodynamically distinct informational filters. Each is hypothesized to possess its own characteristic relaxation time (τ_{res}) and critical activation barrier (Z_c), effectively dividing the thermodynamic load of the thermal gradient. Wild-type measurements provide baseline characterization, but cannot cleanly isolate each stator's intrinsic phase space.

Full experimental validation requires Δ*motCD* and Δ*motAB* isogenic deletion mutants measured across the full thermal gradient to map their independent stability thresholds. Furthermore, a Δ*ycgR* background is strictly required. YcgR is a c-di-GMP-dependent protein that physically interacts with the motor; failing to eliminate this allosteric intervention would artificially alter the friction coefficient and distort the intrinsic Z accumulation, confounding the temperature-dependent speed measurements.

These strains are in preparation. The theoretical framework presented here is therefore offered to the artificial life community as a rigorous, falsifiable prior—a predictive architectural blueprint of biological agency at the thermodynamic limit, rather than a post-hoc rationalization of existing data.

IV. COMPUTATIONAL VALIDATION: THE ZURVATRONIC PHASE DIAGRAM

To bridge the single-molecule *stick-slip* dynamics with population-level stability, a stochastic simulation was implemented in Wolfram Mathematica (v14.1) modeling the accumulated action $Z(\omega,T)$ as a function of rotational speed (ω) across three thermal regimes. Crucially, rotational speed dictates the proton residence time ($\tau_{res} \propto \omega^{-1}$); thus, increasing speed directly restricts the temporal window available for informational computation.The temperature-dependent filtering precision is modeled as:

$$F_{\mathrm{P}}(T) = H_{\max} \exp\left[-\left(\frac{T}{T_0}\right)^2\right] \tag{8}$$

Yielding the Latitudo: $L(T) = k_{\mathrm{B}} T \ln(2) F_{\mathrm{P}}(T)$. Consequently, the probability of a stator unit clearing the informational activation barrier and maintaining its structural agency follows the sigmoidal function:

$$P_{\mathrm{assembly}}(\tau, T) = \frac{1}{1 + \exp\left[-\alpha\left(Z(\tau, T) - Z_c\right)\right]} \tag{9}$$

Where α is a steepness parameter calibrated against literature stator dynamics (Tipping et al., 2013).

Figure A: Phase Diagram

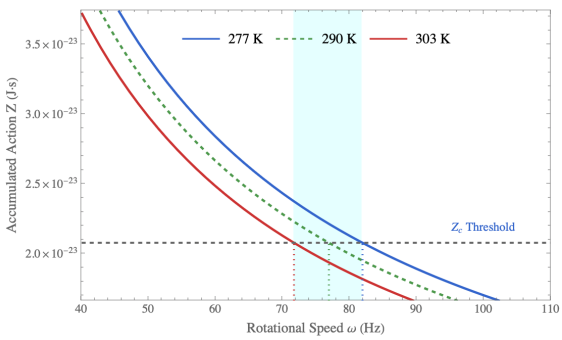

Figure A (Phase Diagram) illustrates Z as a function of rotational speed for 277 K (blue), 290 K (green, dashed), and 303 K (red). The horizontal dashed line marks the critical threshold Zc. The cyan shaded region between the 277 K and 303 K curves defines the *Zurvān Gap*—the specific kinematic interval within which the cold motor retains sufficient filtering capacity to clear the Z_c barrier, while the warm motor has already suffered informational collapse. The vertical dotted lines mark the precise collapse speed ωc for each temperature.

Figure B: Population Disassembly

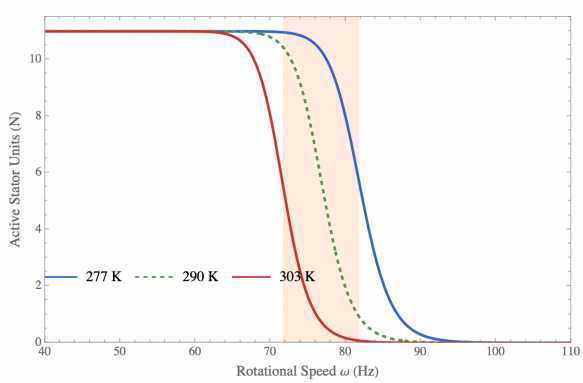

Figure B (Population Disassembly) translates this single-molecule probability into the macroscopic Immanence Coefficient (ζ). It plots the ensemble stator count $N(\omega,T)$ on a linear y-axis (0–11 stator units) against rotational speed. The sigmoidal transition from full engagement ($\zeta \geq 1$) to complete disassembly ($\zeta \to 0$) is clearly visible. The 277 K curve maintains structural integrity at significantly higher speeds than the 303 K curve. The orange shaded region marks the operational *Zurvān Gap*—the speed range where 277 K maintains a near-full stator complement while 303 K has substantially disassembled.

This constitutes a direct, falsifiable experimental prediction: within this high-speed (*near-zero load*) regime, cold-temperature motors should exhibit a statistically higher occupancy of active stator units, potentially measurable via fluorescence imaging of GFP-tagged MotB.

The sensitivity of the collapse speeds to the conformational temperature scale T0 was systematically evaluated. For T0 =260 K, collapse speeds of 81.9 Hz (277 K), 76.9 Hz (290 K), and 71.7 Hz (303 K) are predicted, yielding a Zurvān Gap of approximately 10 Hz. This gap is empirically testable as a measurable separation in the "knee speeds" between cold and warm torque-speed curves.

V. DISCUSSION

V.1 Agency as Informational Silence

The Zurvatronic framework developed here represents a fundamental reorientation in how biological agency is understood at the molecular scale. Rather than asking how a biomolecule accumulates sufficient energy to perform work, it is asked how it achieves sufficient silence—how it prunes the stochastic noise of its thermal environment to allow one functional trajectory to persist where billions of non-functional ones are suppressed.

The key insight is that the cold *P. putida* motor does not fight its thermal environment. It exploits the low-stochasticity floor of cold conditions to achieve filtering precision that warm environments cannot sustain. Life, in this reading, does not resist entropy. It computes against it. In this framework, the molecular machine does not compute in order to move. It computes to persist. The macroscopic mechanical work observed—the directed rotation of the flagellum—is merely the physical byproduct of this continuous informational inhibition.

This shift in ontology is mathematically formalized by the role of κ, the phenomenological elastic coupling constant ($N{\cdot}J{-}1{\cdot}s{-}1$) which mediates the translation of accumulated informational action into physical torque. While κ dictates the absolute magnitude of the classical restorative force, it completely cancels out in the dimensionless Immanence Coefficient (ζ). Consequently, the stability thresholds and the *Zurvān Gap* are invariant to its precise value, provided $\kappa > 0$. The motor's survival does not depend on the strength of its mechanical spring; it depends entirely on the precision of its informational silence.

V.2 Implications for Synthetic Life

Current synthetic biology approaches to engineering molecular motors typically attempt to maximize structural rigidity, operating under the classical intuition that stiffness equates to reliable force generation. This macro-engineering approach consistently produces brittle architectures that fail catastrophically under nanoscale thermal noise. Zurvatronics suggests the exact opposite design philosophy: **subtractive design**. Rather than engineering rigid structures, synthetic biologists should aim for flexible, high-entropy topologies with the thermodynamic capacity to actively compute and suppress

their own decoherence. The ultimate design target for a synthetic *biomotor* is not its mechanical tensile strength, but the precise calibration of its thermal collapse threshold (Z_c) within the Generalized Langevin Equation:

$$m\frac{dv}{dt} = F_{\mathrm{PMF}} - \eta v - \left[P_{\mathrm{assembly}}(\tau, T)\cdot\left(\kappa\cdot Z(t)\cdot\tau_{\mathrm{res}}\right)\right] + \xi(t) \tag{10}$$

By engineering intrinsically disordered regions (IDRs) to undergo a viscoelastic phase transition exactly when $Z(t) \geq Z_c$, synthetic biologists can recreate the *stick-slip* power stroke. The flagellar motor, under this interpretation, is not a fixed mechanical device; it is a continuously self-stabilizing informational system whose identity as a functional agent is renewed at every proton transit. If the Zurvatronic predictions are experimentally confirmed in *P. putida*, this principle becomes directly transferable to the engineering of autonomous synthetic machines that adapt their stability dynamically to extreme environments.

V.3 CONCLUSIONS

Zurvatronics has been introduced as a subtractive theoretical framework for biological agency, grounded in a dimensionally consistent mathematical formalism that unifies Shannon information theory, Landauer's erasure limit, and the medieval intensity calculus of Swineshead's *latitudo formarum*. The framework proposes that agency emerges not from the addition of energy, but from the topological pruning of non-functional conformational trajectories.

Applied to the MotAB flagellar stator, Zurvatronics generates four falsifiable predictions:

1. The Swineshead Latitudo *L(T)* peaks within the psychrotrophic range, creating a predictable *Zurvān Gap* between cold and warm collapse speeds.
2. The *near-zero load* regime is a fundamental singularity of informational instability ($\zeta \approx 1$), resolving the Nirody-Wang paradox. Conflicting experimental observations of duty ratios are the expected macrodynamics of a system hovering on its activation barrier Zc .
3. The MotB 72-100 IDR functions as a Functional Stochastic Manifold, extending the effective proton residence time (τ_{res}) to prevent premature informational collapse at high rotational speeds.
4. Stator disassembly at low load is the population-level signature of the accumulated action failing to reach the critical threshold ($Z < Z_c$).

Zurvatronics is offered to the artificial life community as a theoretical instrument: a framework with sufficient mathematical precision to generate numerical predictions, sufficient philosophical coherence to reframe existing controversies, and sufficient generality to serve as a design blueprint for synthetic systems operating at the thermodynamic edges of viability. The stator that survives the cold does not fight entropy; it computes against it, one proton at a time.

While this study focuses on the MotAB stator of *P. putida*, the Zurvatronic framework points toward a more universal mechanism for biomolecular processing. It is proposed that the complex conformational topology of biological macromolecules serves as an energetic and functional map—an informational operator that earns agency through the systematic inhibition of decoherence. Under this view, the flagellar motor is not a mechanical exception, but a macroscopic window into a fundamental principle of life: molecules do not merely possess structure; they perform a subtractive computation that allows function to persist at the thermodynamic edge.

References

Agmon, N. (1995). The Grotthuss mechanism. *Chemical Physics Letters*, 244(5-6), 456-462.

Berry, R.M. & Berg, H.C. (1997). Absence of a barrier to backwards rotation of the bacterial flagellar motor demonstrated with optical tweezers. *PNAS*, 94(26), 14433-14437.

Bérut, A., et al. (2012). Experimental verification of Landauer's principle linking information and thermodynamics. *Nature*, 483(7388), 187-189.

Castillo, D.J., et al. (2013). The C-terminal periplasmic domain of MotB is responsible for load-dependent control of the number of stators of the bacterial flagellar motor. *Biophysics (Japan)*, 9, 173-181.

Chen, X. & Berg, H.C. (2000). Torque-speed relationship of the flagellar rotary motor of Escherichia coli. *Biophysical Journal*, 78(3), 1036-1041.

Eldar, A. & Elowitz, M.B. (2010). Functional roles for noise in genetic circuits. *Nature*, 467(7312), 167-173.

Engel, G.S., et al. (2007). Evidence for wavelike energy transfer through quantum coherence in photosynthetic systems. *Nature*, 446(7137), 782-786.

Hopfield, J.J. (1974). Kinetic proofreading: A new mechanism for reducing errors in biosynthetic processes requiring high specificity. *PNAS*, 71(10), 4135-4139.

Kauffman, S. (2002). *Investigations*. Oxford University Press.

Kojima, S. & Blair, D.F. (2001). Conformational change in the stator of the bacterial flagellar motor. *Biochemistry*, 40(43), 13041-13050.

Lambert, N., et al. (2013). Quantum biology. *Nature Physics*, 9(1), 10-18.

Nakamura, S., et al. (2020). Direct observation of speed fluctuations of flagellar motor rotation at extremely low load close to zero. *Molecular Microbiology*, 113(4), 755-765.

Nirody, J.A., et al. (2017). Speed of the bacterial flagellar motor near zero load depends on the number of stator units. *PNAS*, 114(47), 12459–12464.

Pfeifer, V., et al. (2022). Role of the two flagellar stators in swimming motility of Pseudomonas putida. *mBio*, 13(4), e02182-22.

Purcell, E.M. (1977). Life at low Reynolds number. *American Journal of Physics*, 45(1), 3-11.

Ramos, J.L., et al. (2010). Pseudomonas putida KT2440: a versatile platform for systems and synthetic biology. *Environmental Microbiology*, 12(2), 442-446.

Reid, S.W., et al. (2006). The maximum number of torque-generating units in the flagellar motor of Escherichia coli is at least 11. *PNAS*, 103(21), 8066-8071.

Ritz, T., et al. (2000). A model for photoreceptor-based magnetoreception in birds. *Biophysical Journal*, 78(2), 707-718.

Swineshead, R. (ca. 1350). *Liber calculationum*. Merton College, Oxford.

Tipping, M.J., et al. (2013). Load-dependent assembly of the bacterial flagellar motor. *mBio*, 4(4), e00551-13.

Wang, B., et al. (2017). Limiting (zero-load) speed of the rotary motor of Escherichia coli is independent of the number of torque-generating units. *PNAS*, 114(47), 12478-12482.

Wraight, C.A. (2006). Chance and design — Proton transfer in water, channels and bioenergetic proteins. *Biochimica et Biophysica Acta*, 1757(8), 886-912.

Zaehner, R.C. (1955). *Zurvan: a Zoroastrian Dilemma*. Clarendon Press.